Proximal approximation of firmly nonexpansive maps with small derivative asymmetry

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Abstract

We study uniform approximation of a firmly nonexpansive map by the proximal map of a single convex function. Let H be a real Hilbert space, let $R > 0$, and suppose that T is continuously Fréchet differentiable on a neighbourhood of the closed ball $B_{\sqrt{2}R}$, firmly nonexpansive on that ball, and satisfies $\sup_{B_{\sqrt{2}R}} \|DT - DT^*\| \le \varepsilon$. We prove that a finite continuous convex function φ exists with $\sup_{B_R} \|T - \mathrm{prox}_\varphi\| \le R\sqrt{2\varepsilon}$. The estimate is independent of dimension and requires no bound on derivatives of the skew part. The proof converts the differential defect into approximate convex interpolation inequalities, then applies an elementary form of established approximate-subdifferential integration. An affine proximal approximation gives the complementary bound $R/2$. For a planar rotation perturbation with $0 \le \varepsilon \le 1$, the exact approximation error is $\varepsilon R/2$; thus this example does not establish optimality of the square-root rate.

Keywords: proximal map; firm nonexpansiveness; approximate subdifferential; convex interpolation; quantitative stability.

1 Introduction

A proximal map associates an input x with the unique minimizer of a convex function penalized by its squared distance from x. For a proper lower-semicontinuous convex function $\varphi : H \to (-\infty, +\infty]$ on a real Hilbert space H, we use the normalization

$$\mathrm{prox}_\varphi(x) = \operatorname*{argmin}_{p \in H} \left\{ \varphi(p) + \frac{1}{2}\|p - x\|^2 \right\}. \tag{1.1}$$

Such maps are firmly nonexpansive. This stability property alone does not imply that a map is proximal: a nonzero rotational component can preserve firm nonexpansiveness while preventing the map from being a gradient. Moreau's characterization identifies convex proximal maps with nonexpansive gradients of convex potentials [1]; differential characterizations and extensions are developed by Gribonval and Nikolova [2]. These exact results motivate a quantitative question: how close is a firmly nonexpansive map to a convex proximal map when its derivative is almost self-adjoint?

Write $B_r = \{x \in H : \|x\| \le r\}$. A map T is firmly nonexpansive on a set D if

$$\|T(x) - T(y)\|^2 \le \langle T(x) - T(y), x - y \rangle \qquad (x, y \in D). \tag{1.2}$$

This condition implies both monotonicity and Lipschitz continuity with constant one. For a bounded linear operator A on H, A^* denotes its Hilbert adjoint, and the norm of an operator is

its operator norm. Throughout the paper, the asymmetry parameter bounds the full difference $DT - DT^*$, without a factor $1/2$.

Theorem 1.1. *Let H be a real Hilbert space, $R > 0$, and $\varepsilon \geq 0$. Suppose that $T : U \to H$ is continuously Fréchet differentiable on an open set U containing $B_{\sqrt{2}R}$, is firmly nonexpansive on $B_{\sqrt{2}R}$, and satisfies*

$$\sup_{x \in B_{\sqrt{2}R}} \|DT(x) - DT(x)^*\| \leq \varepsilon. \tag{1.3}$$

Then there is a finite continuous convex function $\varphi : H \to \mathbb{R}$ such that

$$\sup_{x \in B_R} \|T(x) - \operatorname{prox}_\varphi(x)\| \leq R\sqrt{2\varepsilon}. \tag{1.4}$$

The same globally defined penalty serves all inputs in B_R. The approximation is local, and the assumptions are imposed on the larger ball because the proof compares values at a shifted point that can lie outside B_R. Neither the dimension of H nor a modulus of continuity for DT enters the bound.

There is also an approximation that requires no derivative information. The affine map $x \mapsto T(0) + x/2$ is proximal and approximates every firmly nonexpansive T on B_R with error at most $R/2$. Proposition 3.1 gives the short proof. Consequently, if $\Gamma_0(H)$ denotes the proper lower-semicontinuous convex functions on H, Theorem 1.1 yields

$$\inf_{\varphi \in \Gamma_0(H)} \sup_{x \in B_R} \|T(x) - \operatorname{prox}_\varphi(x)\| \leq R \min\left\{\frac{1}{2},\, \sqrt{2\varepsilon}\right\}. \tag{1.5}$$

The derivative-based estimate improves the affine guarantee when $\varepsilon < 1/8$, and tends to zero with the asymmetry.

Relation to earlier results. The convex-integration step of the proof is established machinery. Marcellin and Thibault [3, Lemma 2.2] characterize families of graphs contained in the approximate subdifferentials of one convex function. Applied to the graph variables $(T(x), x - T(x))$, their lemma supplies the fixed-error proximal approximation used below. Our contribution is to derive its compatibility condition from (1.2) and (1.3), with error $2\varepsilon R^2$, and thereby obtain (1.4). We give a direct affine-support construction to make the resulting penalty and constants explicit.

The passage from differential information to interpolation inequalities adapts the shifted-gradient comparison used in smooth convex analysis. Drori [4, Theorem 3.1] proves a local form of that comparison and also gives a counterexample to unrestricted interpolation over an open convex domain [4, Proposition 2.1]. Here firm nonexpansiveness controls both the location of the shifted point and the accumulated error of the radial potential. Convex-envelope formulas of Azagra, Le Gruyer and Mudarra [5] provide another route to the repair, while the quadratic-conjugate construction in Zoll and Grimmer [6, Theorem 3.3 and its proof] has the same underlying dual form. The fixed-error conversion is not a new extension theorem. These envelope constructions begin with scalar values and vector data satisfying interpolation inequalities. Lemma 2.2 derives the fixed-error inequality needed here from firm nonexpansiveness and the derivative bound on T.

Lauga and Vaiter [7] develop characterizations of inexact proximal operators relative to a specified reference function. Their Proposition 9 gives the approximate-subgradient residual estimate used at the end of our proof, in a finite-dimensional weakly convex setting. Theorem 1.1 begins with a derivative defect of T, without a reference penalty, and constructs one convex penalty that works throughout the smaller ball. The result is an existence and stability estimate; it does not provide a complexity bound for computing that penalty or its proximal map.

2 From derivative asymmetry to a convex penalty

The proof has two tasks. We first integrate T radially and estimate how far the resulting scalar function is from having T as its gradient. Firm nonexpansiveness then turns that error into an approximate convex interpolation inequality. A dual change of variables converts this inequality into approximate supporting planes of the desired penalty.

2.1 Radial integration and the shifted-point estimate

Define

$$f(x) = \int_0^1 \langle T(tx), x\rangle \, dt \qquad (x \in B_{\sqrt{2}R}). \tag{2.1}$$

This is the classical radial homotopy construction; see, for example, the discussion of Poincaré integral operators in [8]. The calculation below records the Hilbert-space form and the normalization needed here.

Lemma 2.1. *Under the hypotheses of Theorem 1.1,*

$$\nabla f(x) - T(x) = \int_0^1 t\big(DT(tx)^* - DT(tx)\big)x \, dt, \qquad \|\nabla f(x) - T(x)\| \le \frac{\varepsilon}{2}\|x\|. \tag{2.2}$$

Proof. For each fixed x, the segment $\{tx : 0 \le t \le 1\}$ is compact and lies in U. It has a neighbourhood on which the radial formula is defined and differentiation under the integral is justified by continuity of DT. Thus

$$\nabla f(x) = \int_0^1 \big(T(tx) + tDT(tx)^* x\big) \, dt.$$

On the other hand, integrating the derivative of $tT(tx)$ gives

$$T(x) = \int_0^1 \big(T(tx) + tDT(tx)x\big) \, dt.$$

Subtraction proves the identity, and (1.3) gives the estimate. This argument uses compactness of the radial segment, not compactness of a Hilbert ball. □

The radial function need not be convex. The next estimate provides the information that will replace exact convexity in the construction of the penalty.

Lemma 2.2. *For $x, y \in B_R$, the function in* (2.1) *satisfies*

$$f(x) \ge f(y) + \langle T(y), x - y\rangle + \frac{1}{2}\|T(x) - T(y)\|^2 - \Delta, \qquad \Delta = 2\varepsilon R^2. \tag{2.3}$$

Proof. Set $d = x - y$, $a = T(x) - T(y)$, and $z = x - a$. Firm nonexpansiveness gives

$$\|a\|^2 + \|d - a\|^2 \le \|d\|^2, \qquad \|a - d/2\| \le \|d\|/2. \tag{2.4}$$

Consequently

$$\|z\| \le \left\|\frac{x+y}{2}\right\| + \frac{\|x-y\|}{2} \le \sqrt{2}R,$$

where the last inequality follows from the parallelogram identity and $\|x\|, \|y\| \le R$. The segments joining x to z and y to z therefore lie in $B_{\sqrt{2}R}$. Figure 1(a) illustrates why a larger observation ball enters.

Put $e(u) = \nabla f(u) - T(u)$. Lemma 2.1 gives $\|e(u)\| \le \eta := \varepsilon R/\sqrt{2}$ on this ball. For $h_y(u) = f(u) - \langle T(y), u\rangle$, integration along $x - ta$ gives

$$\begin{aligned} h_y(z) - h_y(x) &= -\int_0^1 \langle T(x - ta) - T(y) + e(x - ta), a\rangle \, dt \\ &\le -\|a\|^2 + \int_0^1 t\|a\|^2 \, dt + \eta\|a\| = -\frac{1}{2}\|a\|^2 + \eta\|a\|. \end{aligned} \tag{2.5}$$

Here Lipschitz continuity bounds $T(x - ta) - T(x)$. Along the other segment, monotonicity gives $\langle T(y + t(z - y)) - T(y), z - y\rangle \ge 0$ for $t > 0$; at $t = 0$ the inner product vanishes. Hence

$$h_y(z) - h_y(y) \ge -\eta\|z - y\|. \tag{2.6}$$

Subtracting (2.5) from (2.6) yields

$$h_y(x) - h_y(y) \ge \frac{1}{2}\|a\|^2 - \eta(\|a\| + \|d - a\|).$$

Finally, (2.4) implies

$$\|a\| + \|d - a\| \le \sqrt{2}\|d\| \le 2\sqrt{2}R.$$

The error is therefore at most $2\varepsilon R^2$, which proves (2.3). □

2.2 Approximate supporting planes

For $h \in \Gamma_0(H)$, $p \in \operatorname{dom} h$, and $\delta \ge 0$, the approximate subdifferential is

$$\partial_\delta h(p) = \{u \in H : h(q) \ge h(p) + \langle u, q - p\rangle - \delta \text{ for every } q \in H\}.$$

For $p \notin \operatorname{dom} h$, set $\partial_\delta h(p) = \varnothing$. Lowering the affine function $q \mapsto h(p) + \langle u, q - p\rangle$ by δ makes it a global minorant of h. The following consequence of approximate convex integration is stated with bounded data because that is all the proof of Theorem 1.1 requires.

Lemma 2.3 (Fixed-error proximal approximation)**.** *Let $E \subset H$ be nonempty and bounded. Let $f : E \to \mathbb{R}$ and $p : E \to H$ be bounded, and suppose that, for some $\Delta \ge 0$,*

$$f(x) \ge f(y) + \langle p(y), x - y\rangle + \frac{1}{2}\|p(x) - p(y)\|^2 - \Delta \qquad (x, y \in E). \tag{2.7}$$

Define

$$\varphi(q) = \sup_{x \in E}\left\{-f(x) + \frac{1}{2}\|p(x)\|^2 + \langle x - p(x), q\rangle\right\}. \tag{2.8}$$

Then φ is finite, continuous and convex on H, and

$$\|p(x) - \operatorname{prox}_\varphi(x)\| \le \sqrt{\Delta} \qquad (x \in E). \tag{2.9}$$

Proof. The proximal optimality condition is $p = \operatorname{prox}_\varphi(x)$ if and only if $x - p \in \partial\varphi(p)$. This motivates approximate supporting planes with slopes $x - p(x)$ at the output points $p(x)$. Set $u_x = x - p(x)$ and $v_x = \langle p(x), x\rangle - f(x) - \|p(x)\|^2/2$. Rearranging (2.7) gives

$$v_y \ge v_x + \langle u_x, p(y) - p(x)\rangle - \Delta. \tag{2.10}$$

Formula (2.8) can be written

$$\varphi(q) = \sup_{x \in E}\{v_x + \langle u_x, q - p(x)\rangle\}.$$

The affine functions have uniformly bounded slopes and intercepts, so their supremum is finite and globally Lipschitz, as well as convex. Taking the supremum in (2.10), and using the term indexed by y for the lower bound, gives

$$v_y \le \varphi(p(y)) \le v_y + \Delta.$$

For every $q \in H$, the affine function indexed by y consequently yields

$$\varphi(q) \ge \varphi(p(y)) + \langle u_y, q - p(y)\rangle - \Delta.$$

Thus $y - p(y) \in \partial_\Delta \varphi(p(y))$.

Write $z = \operatorname{prox}_\varphi(y)$, $p = p(y)$, and $Q_y(q) = \varphi(q) + \|q - y\|^2/2$. Adding the quadratic term to the last inequality gives

$$Q_y(z) \ge Q_y(p) + \frac{1}{2}\|z - p\|^2 - \Delta.$$

The function Q_y is 1-strongly convex, so its minimizer also satisfies

$$Q_y(p) \ge Q_y(z) + \frac{1}{2}\|p - z\|^2.$$

Combining these two inequalities gives $\|p - z\|^2 \le \Delta$, proving (2.9). Equivalently, this is the approximate-monotonicity estimate obtained by pairing $y - p \in \partial_\Delta \varphi(p)$ with $y - z \in \partial\varphi(z)$; compare [7, Proposition 9]. □

Proof of Theorem 1.1. Take $E = B_R$, $p(x) = T(x)$, and f from (2.1). The data are bounded:

$$\|T(x)\| \le \|T(0)\| + R, \qquad |f(x)| \le R\|T(0)\| + \frac{R^2}{2}.$$

Lemma 2.2 supplies (2.7) with $\Delta = 2\varepsilon R^2$. Lemma 2.3 gives a finite continuous convex φ on all of H, and (2.9) becomes (1.4). □

To relate this direct proof explicitly to Marcellin–Thibault [3, Lemma 2.2], define

$$A_\delta(q) = \begin{cases} \{x - p(x) : x \in E,\ p(x) = q\}, & \delta \ge \Delta, \\ \varnothing, & 0 \le \delta < \Delta. \end{cases}$$

Summing (2.10) around a cycle of n selected graph points gives $\sum_i \langle u_i, p_i - p_{i+1}\rangle \ge -n\Delta \ge -\sum_i \delta_i$, with cyclic indices. This is precisely their cyclic-family condition. Their lemma gives a convex function containing each A_δ in its δ-subdifferential. Empty levels below Δ are allowed; neither maximality nor injectivity of p is required. Formula (2.8) gives a direct construction when the scalar data $f(x)$ are available.

The dual-support formula also identifies the relation with the quadratic-envelope route. If

$$q_x(w) = f(x) + \langle p(x), w - x\rangle + \frac{1}{2}\|w - x\|^2,$$

then completing the square in the Fenchel conjugate gives

$$q_x^*(q) = -f(x) + \langle x, q\rangle + \frac{1}{2}\|q - p(x)\|^2, \qquad \varphi(q) = \sup_{x \in E} q_x^*(q) - \frac{1}{2}\|q\|^2.$$

This is the same dual construction underlying the envelope comparisons in [5, 6]. The direct proof above avoids a separate smooth-envelope regularity theorem.

3 The affine baseline and an exact planar example

The estimate in Theorem 1.1 becomes informative as derivative asymmetry decreases. The comparison in (1.5) follows from a proximal approximation available without that information.

Proposition 3.1. *If T is firmly nonexpansive on B_R, then the convex function*

$$\varphi_{\mathrm{aff}}(p) = \frac{1}{2}\|p\|^2 - 2\langle T(0), p\rangle$$

satisfies

$$\mathrm{prox}_{\varphi_{\mathrm{aff}}}(x) = T(0) + \frac{1}{2}x, \qquad \sup_{x\in B_R} \|T(x) - \mathrm{prox}_{\varphi_{\mathrm{aff}}}(x)\| \le \frac{R}{2}.$$

Proof. The proximal formula follows by minimizing the quadratic objective. If $a = T(x) - T(0)$, firm nonexpansiveness gives

$$\|a - x/2\|^2 = \|a\|^2 - \langle a, x\rangle + \frac{1}{4}\|x\|^2 \le \frac{1}{4}\|x\|^2.$$

□

The next family has an exactly computable distance from the convex proximal maps. It shows that a positive derivative defect can force a positive approximation error, and that the affine constant $1/2$ is sharp over the unrestricted firmly nonexpansive class in the plane.

Proposition 3.2. *On $\mathbb{R}^2$, let $J(x_1, x_2) = (-x_2, x_1)$ and $T_\varepsilon = (I + \varepsilon J)/2$, where $0 \le \varepsilon \le 1$. Then T_ε is firmly nonexpansive, $\|DT_\varepsilon - DT_\varepsilon^*\| = \varepsilon$, and*

$$\inf_{\varphi\in\Gamma_0(\mathbb{R}^2)} \sup_{x\in B_R} \|T_\varepsilon(x) - \mathrm{prox}_\varphi(x)\| = \frac{\varepsilon R}{2}. \tag{3.1}$$

Proof. For $d \in \mathbb{R}^2$,

$$\|T_\varepsilon d\|^2 = \frac{1+\varepsilon^2}{4}\|d\|^2 \le \frac{1}{2}\|d\|^2 = \langle T_\varepsilon d, d\rangle.$$

Also $DT_\varepsilon - DT_\varepsilon^* = \varepsilon J$. The proximal map $P_0(x) = x/2$, corresponding to $\varphi_0(p) = \|p\|^2/2$, gives the upper bound in (3.1).

For the lower bound, orient the circle C_R counterclockwise. Every convex proximal map P is the gradient of a continuously differentiable potential on $\mathbb{R}^2$ [1], so its circulation around C_R is zero. Direct parametrization of the circle gives

$$\int_{C_R} \langle T_\varepsilon(x), dx\rangle = \varepsilon\pi R^2.$$

If $e = \sup_{B_R} \|T_\varepsilon - P\|$, the circle's length therefore gives

$$\varepsilon\pi R^2 = \left|\int_{C_R} \langle T_\varepsilon(x) - P(x), dx\rangle\right| \le 2\pi R e.$$

Thus $e \ge \varepsilon R/2$. □

Figure 1(b) separates the general guarantee from the exact planar calculation. The linear error of the rotation family does not establish optimality of the square-root rate in Theorem 1.1. In particular, the present argument does not decide whether a smaller asymptotic upper bound holds for the whole class.

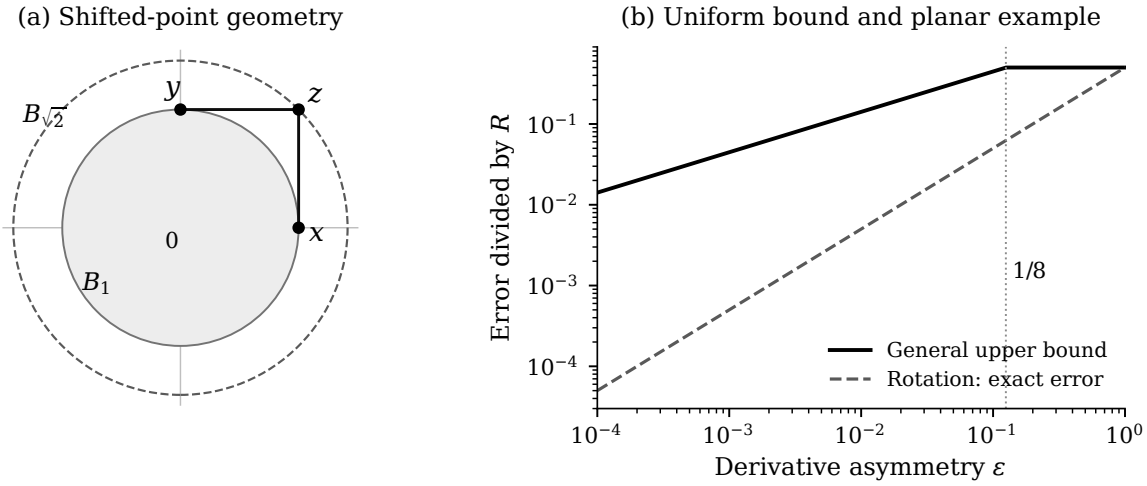


Figure 1: The observation region and the approximation estimates. (a) With $R = 1$, the shifted point $z = x - (T(x) - T(y))$ can reach the boundary of $B_{\sqrt{2}}$: take $x = (1, 0)$, $y = (0, 1)$, and $T = (I - J)/2$, giving $z = (1, 1)$. The shaded disk is B_1; the dashed circle is $B_{\sqrt{2}}$. This illustrates the geometry of the proof, not optimality of the theorem's observation radius. (b) The general upper bound $\min\{1/2, \sqrt{2\varepsilon}\}$ for normalized error is shown with the exact value $\varepsilon/2$ for the planar family in Proposition 3.2. The latter is an example-specific value, not the general approximation error. The vertical line marks $\varepsilon = 1/8$.

4 Scope of the approximation

The derivative hypothesis provides a uniform certificate of proximity to one convex optimization model. At $\varepsilon = 0$, the constructed proximal map agrees with T throughout B_R; for positive ε, the error remains uniform over that ball and independent of dimension. The radius $\sqrt{2}R$ suffices because of the shifted-point geometry in Lemma 2.2. We do not claim that this radius is minimal for the theorem itself.

Formula (2.8) is explicit as a supremum over the available field data, but evaluating that supremum and its proximal minimizer may require further structure. The theorem supplies neither a finite-data certification procedure nor a computational complexity bound. Its approximation estimate also concerns maps on a prescribed region; convergence claims for iterations involving the original map require a separate argument.

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Data and code. The results are analytic. The accompanying source package contains the figure-generation script; the plots illustrate the proved geometry and formulas and are not used as proof evidence.

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